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PCC-203-COM: Operating Systems

Unit 2:Process and Thread Management

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Course Objectives

1. To understand the fundamental concepts, types, and structures of Operating Systems.
2. To understand and analyze process management concepts.
3. To identify and solve concurrency and deadlock in the operating system.
4. Explore the various techniques of memory management.
5. Understand I/O management, disk scheduling, and file systems.

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Course Outcomes

Upon successful completion of this course, students will be able to:

CO1: **Analyze** the fundamentals of Operating Systems, including types, structures, system calls,and basic Linux commands.

CO2: **Apply** process scheduling and synchronization to optimize CPU utilization in modern operating systems.

CO3: **Identify** the mechanism for dealing with deadlocks and concurrency concerns.

CO4: **Apply** techniques of memory management to solve memory management problems

CO5: **Illustrate** I/O and file management policies.

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Contents

- **Process management:** Definition,
- Types of process.
- Process States and Transitions diagram,
- Process Control Block (PCB),
- context switching and its impact on performance,
- Process Scheduling.
- **Types of Schedulers:** long term, short term, middle term,
- Threads: Concept of thread,
- Multithreading,
- User-level vs Kernel-level Threads.
- **Scheduling Algorithms:** Preemptive Scheduling vs Non-preemptive Scheduling,
- FCFS, SJF, RR,
- Priority Process Scheduling in UNIX and Windows
- **Cast Studies** - 1) Prepare case study on challenges for Real-time Scheduling 2)Performance Comparison of Scheduling Algorithms (FCFS vs SJF, RR vs Priority Scheduling) under different workload conditions

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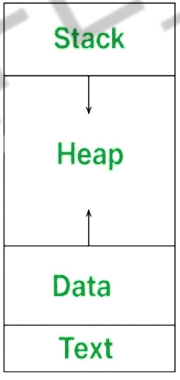
Process In Operating System

- A process is a program in execution. For example, when we write a program in C or C++ and compile it, the compiler creates binary code. The original code and binary code are both programs. When we actually run the binary code, it becomes a process.
- A process is an 'active' entity instead of a program, which is considered a 'passive' entity.
- A single program can create many processes when run multiple times; for example, when we open a .exe or binary file multiple times, multiple instances begin (multiple processes are created). .

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Process In Memory

- A process in memory is divided into several distinct sections, each serving a different purpose. Here's how a process typically looks in memory.



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Process In Memory

- **Text Section**: A text or code segment contains executable instructions. It is typically a read only section
- **Stack**: The stack contains temporary data, such as function parameters, returns addresses, and local variables.
- **Data Section**: Contains the global variable.
- **Heap Section**: Dynamically memory allocated to process during its run time.

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Process Management

- Process Management for a single tasking or batch processing system is easy as only one process is active at a time.
- With multiple processes (multiprogramming or multitasking) being active, the process management becomes complex as a CPU needs to be efficiently utilized by multiple processes.
- Multiple active processes can may share resources like memory and may communicate with each other.
- This further makes things complex as an Operating System has to do process synchronization.
- The main advantages of having multiprogramming are system responsiveness and better CPU utilization.
- We can run multiple processes in interleaved manner on a single CPU. For example, when the current process is getting busy with IO, we assign CPU to some other

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CPU Bound v/s I/O Bound Processes

- A CPU-bound process requires more CPU time or spends more time in the running state. An I/O-bound process requires more I/O time and less CPU time. An I/O-bound process spends more time in the waiting state.
- Process planning is an integral part of the process management operating system. It refers to the mechanism used by the operating system to determine which process to run next. The goal of process scheduling is to improve overall system performance by maximizing CPU utilization, minimizing throughput time, and improving system response time.

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Process Management Taks

- Process Creation and Termination : Process creation involves creating a Process ID, setting up Process Control Block, etc. A process can be terminated either by the operating system or by the parent process. Process termination involves clearing all resources allocated to it.
- CPU Scheduling : In a multiprogramming system, multiple processes need to get the CPU. It is the job of Operating System to ensure smooth and efficient execution of multiple processes.
- Deadlock Handling : Making sure that the system does not reach a state where two or more processes cannot proceed due to cyclic dependency on each other.
- Inter-Process Communication : Operating System provides facilities such as shared memory and message passing for cooperating processes to communicate.
- Process Synchronization : Process Synchronization is the coordination of execution of multiple processes in a multiprogramming system to ensure that they access shared resources (like memory) in a controlled and predictable manner.

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Process States

- As a process executes, it changes **state**. The state of a process is defined in part by the current activity of that process. A process may be in one of the following states:
- **New**. The process is being created.
- **Running**. Instructions are being executed.
- **Waiting**. The process is waiting for some event to occur (such as an I/O completion or reception of a signal).
- **Ready**. The process is waiting to be assigned to a processor.
- **Terminated**. The process has finished execution.

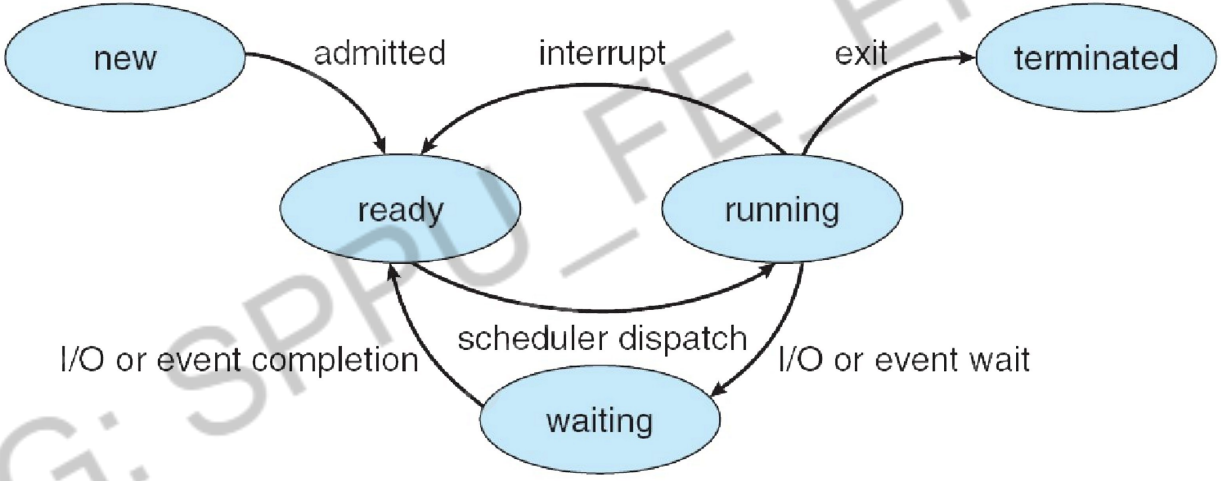
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Process States

- These names are arbitrary, and they vary across operating systems. The states that they represent are found on all systems, however.
- Certain operating systems also more finely delineate process states. It is important to realize that only one process can be *running* on any processor at any instant. Many processes may be *ready* and *waiting,* however.

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Process State Transition Diagram



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Process Control Block

- Each process is represented in the operating system by a **process control block (PCB)**—alsocalleda **task control block**.]
- It contains many pieces of information associated with a specific process, including these:

✔ **Process state**. The statemay be new, ready, running, waiting, halted, and so on.

✔ **Program counter**. The counter indicates the address of the next instruction to be executed for this process.

✔ **CPU registers**. The registers vary in number and type, depending on the computer architecture. They include accumulators, index registers, stack pointers, and general-purpose registers, plus any condition-code information. Along with the program counter, this state information must be saved when an interrupt occurs, to allow the process to be continued correctly afterward.

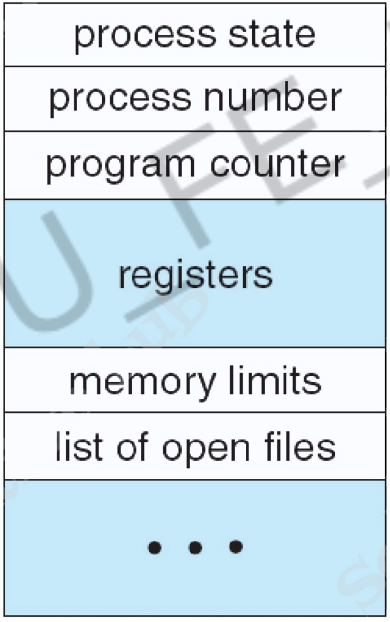
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Process Control Block

- **CPU-scheduling information**. This informationincludes aprocesspriority, pointers to scheduling queues, and any other scheduling parameters.
- **Memory-management information**. This information may include such items as the value of the base and limit registers and the page tables, or the segment tables, depending on the memory system used by the operating system
- **Accounting information**. This information includes the amount of CPU and real time used, time limits, account numbers, job or process numbers, and so on.
- **I/O status information**. This information includes the list of I/O devices allocated to the process, a list of open files, and so on. In brief, the PCB simply serves as the repository for any information that may vary from process to process.

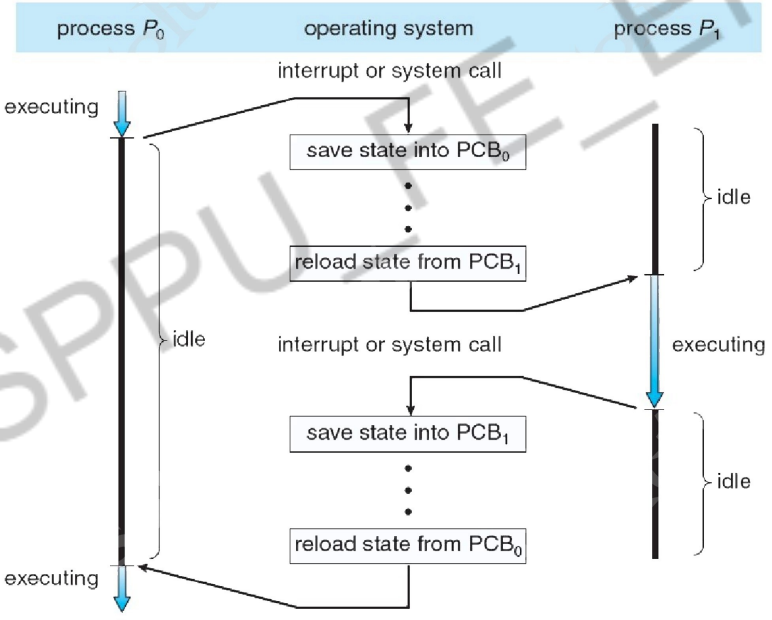
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Process Control Block(PCB)



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Process Control Block(PCB)



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Context Switching

- Context Switching in an operating system is a critical function that allows the CPU to efficiently manage multiple processes.
- The ability for a process to be switched from the CPU so that another process can run is known as context switching.
- By saving the state of a currently active process and loading the state of another, the system can handle various tasks simultaneously without losing progress.
- This switching mechanism ensures optimal use of the CPU, enhancing the system's ability to perform multitasking effectively.
- In the context switching of two processes, the priority-based process occurs in the ready queue of the process control block. Following are the steps:

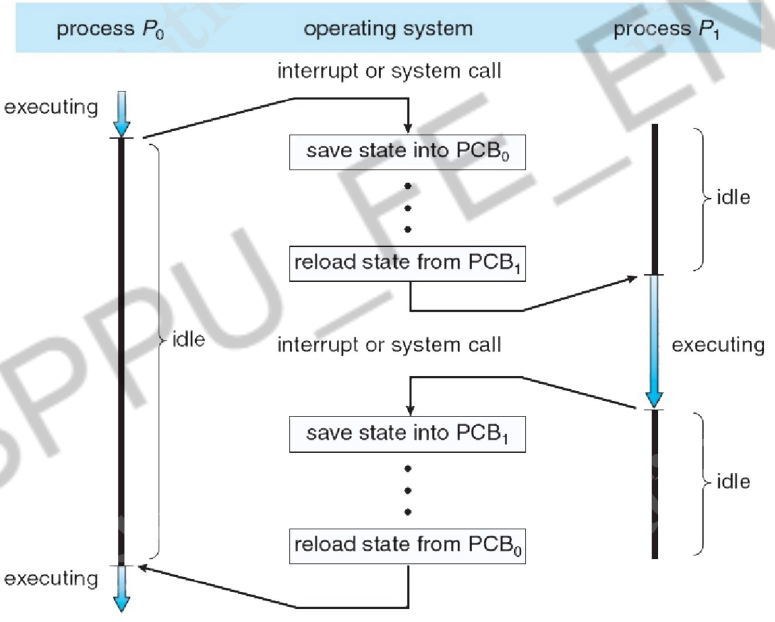
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Context Switching

- The state of the current process must be saved for rescheduling.
- The process state contains records, credentials, and operating system-specific information stored on the PCB or switch.
- The PCB can be stored in a single layer in kernel memory or in a custom OS file.
- A handle has been added to the PCB to have the system ready to run.
- The operating system aborts the execution of the current process and selects a process from the waiting list by tuning its PCB.
- Load the PCB's program counter and continue execution in the selected process.
- Process/thread values can affect which processes are selected from the queue, this can be important.

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Context Switching



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